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FACULTY OF COMPUTING
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A3: Agent Ideation and Formulation

Faculty of Computing

Project <Intelligent Volunteer Matching>

Team: <Visionary.AI>

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1. PEAS Model Representation

The proposed agent is an Intelligent Volunteer Matching Agent designed to automate volunteers' recommendations for available volunteering opportunities. The agent analyses volunteer profiles and opportunity requirements to make sure that assignments are appropriate, efficient, and reliable. Its main goal is to make the matching process better by considering a certain criterion such as skills, availability, location, interests, and reliability, while at the same time minimizing time conflicts and bad matches. The agent helps event organizers by reducing manual effort and improving the quality of volunteer selection.

Table 1.0

PEAS Elements	Describe the components in your agent based on PEAS
PERFORMANCE	Is how success is measured it will be through these standards: ● Match Quality scores (percent of assignments that satisfy high match) ● Fill rate: How many volunteers match conditions. ● Volunteer Reliability Impact: reduction of no-shows/cancellations.
ENVIRONMENT	Where and who will use the agent: ● University/NGO volunteer program context. ● Data sources: volunteer profiles, opportunity postings, schedules, locations, cause categories. ● Constraints: limited slots, time windows, distance limits, minimum reliability.

	● Stakeholders:volunteers,organizers/coordinators, admin staff.
ACTUATORS	It is what the agent can do. ● Recommend / rank volunteers for each opportunity. ● Notify organizers with top candidates. ● Notify volunteers of suggested opportunities. ● Create/Update assignments in a database ● Reject/prune invalid matches automatically when a KR check fails
SENSORS	It is what agent Perceives and reads ● Volunteer input like skills & levels, availability time slots, location, interests, reliability score. ● Opportunity input from stakeholders like tasks required skills, time slots, location and needed headcount. ● Historical logs like attendance history.

Short Explanation:

Within the PEAS framework, the Intelligent Volunteer Matching Agent makes decisions as a rational agent. Performance measures are used to ensure that the agent makes excellent volunteer matches that are free of scheduling conflicts and that the reliability of the volunteers increases. The environment is characterized by volunteer programs that involve multiple users, opportunities, and constraints simultaneously. Sensors are used by the agent to receive information on volunteer and job postings, including skills, volunteer times, volunteer locations, volunteer interests, and past reliability. Sensors and actuators work together within the PEAS

framework and enable the Intelligent Volunteer Matching Agent to make decisions that meet its purpose by matching volunteer skills with organizational requirements.

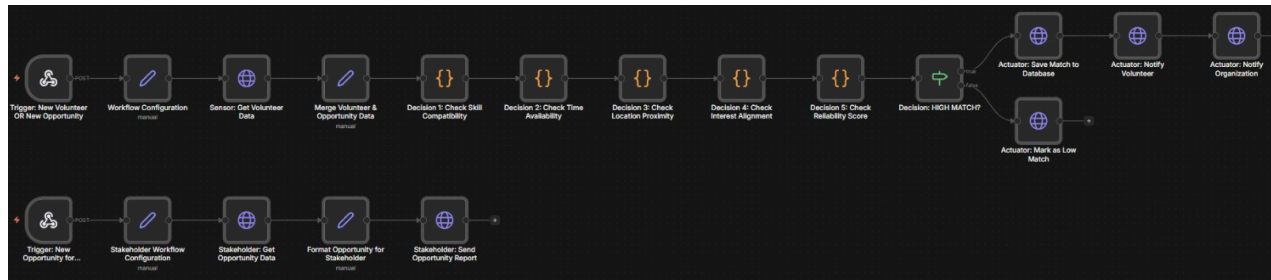


Figure 1 Agent Automated Workflow

Short Explanation:

This automation does the heavy lifting of connecting the right people with the right causes by running two synchronized processes.

First, it handles the Matching Workflow: whenever a new volunteer signs up or a project is posted, the system instantly vets the pair across five key areas — skills, availability, location, interests, and reliability. If it's a "High Match," the system automatically logs the data and sends immediate notifications to both the volunteer and the organization so they can get started right away.

Second, we've added a Stakeholder Reporting track. This ensures that every time a new opportunity is created, the system automatically formats the details into a professional report and sends it directly to management. This keeps the whole team informed in real-time without anyone having to manually copy and paste data into emails.

2. Agent Type

The Intelligent Volunteer Matching System employs a goal-based agent. This specific agent type is the most suitable choice for our requirements. A goal-based agent operates by defining a clear objective. It selects sequences of actions to achieve this goal. It does not merely react to current inputs like a reflex agent.

The system focuses on a primary goal. We aim to create high-quality matches between volunteers and opportunities. The agent actively avoids time clashes. It also minimizes travel distances and filters out unreliable volunteers.

We define the desired goal state using the PEAS model. The objective is to fill all open opportunities. Selected volunteers must satisfy specific constraints. These constraints include skills, availability, and location. Interest and reliability scores are also critical factors.

The agent utilizes sensors to perceive data. It reads volunteer profiles and opportunity details. Figure 1 illustrates the agent's internal model. This workflow dictates how specific actions change the system state. These actions include checking skills, verifying availability, and classifying matches. Other actions involve notifying users and saving data to the database.

The system triggers the agent when a new volunteer or opportunity appears. The agent evaluates the candidates immediately. It only executes actions that advance the system toward the goal state. For example, it recommends a volunteer if all conditions are met. Conversely, it rejects a candidate if any constraint fails.

We compared this approach with a simple reflex agent. A reflex agent relies on isolated IF-THEN rules. In contrast, our goal-based agent reasons explicitly about the final objective. This agent also differs from a utility-based agent. It does not need to compute complex numerical trade-offs. The system simply determines if a "good match" exists. This distinction makes the goal-based agent a realistic choice. It is simple to implement for this project. However, it remains sufficiently intelligent to ensure accurate matching within our automated workflow.

3. Proof of Concept Description

The proposed proof of concept demonstrates an automated, goal-based workflow that instantly processes new volunteer registrations and opportunity postings to identify optimal pairings. As depicted in Figure 1, the system utilizes a sequential filtering mechanism where every potential match must successfully pass through five distinct decision nodes: Skill Compatibility, Time Availability, Location Proximity, Interest Alignment, and Reliability Scoring. This design directly applies the logic defined in our Knowledge Representation to ensure the agent only promotes "High Match" candidates who satisfy every operational constraint, thereby eliminating the inefficiencies and errors common in manual screening.

Once a valid match is confirmed, the workflow automatically triggers specific actuators to finalize the process without human intervention. The system logs the assignment in the database and instantly dispatches notification emails to both the volunteer and the organization to facilitate rapid coordination. Additionally, the functionality includes a parallel stakeholder reporting track that formats and sends professional summaries of new opportunities to management. This end-to-end automation serves as a functional validation that the agent can autonomously achieve its goal of increasing high-quality volunteer engagement while significantly reducing administrative overhead.

4. References

Saci, S. A. (2025, May 10). *Automate supply chain analytics workflows with AI agents using n8n. Towards Data Science.*

<https://towardsdatascience.com/automate-supply-chain-analytics-workflows-with-ai-agents-using-n8n/> (Accessed December 22, 2025)